

RISK LOG v1.0 - Team 05

CA4 Critical Design Review (CDR) MC3113 Mechatronic Systems Design

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- CA1 Standard: ISO 9241-210 (Human-Centred Design)

1. RISK SUMMARY TABLE

Risk ID	Description	Likelihood	Impact	Mitigation Plan	Owner	Status
RISK-01	Time exceeds 75s	Medium (was)	High	Increased v_base from 0.45 to 0.52 m/s	Member 180	 CLOSED
RISK-02	IAE exceeds 0.025	Medium (was)	High	Fine-tuned Kp=2.3, Kd=0.13, added anti-windup	Member 180	 CLOSED
RISK-03	Energy exceeds 60	Low	Medium	Optimized v_base and gains	Member 180	 CLOSED
RISK-04	S2 obstacle fails	Medium (was)	High	Increased threshold to 1.2m, reduced speed to 0.35 m/s	Member 180	 CLOSED
RISK-05	S3 fault recovery fails	Medium (was)	High	Implemented state machine with 2s validation	Member 180	 CLOSED

RISK-06	IAE at 0% margin (0.025)	Low	Medium	Improved IAE to 0.021 (4.5% margin)	Member 180	 CLOSED
RISK-07	Time at 0% margin (75.0s)	Low	Medium	Improved time to 71.5s (4.7% margin)	Member 180	 CLOSED
RISK-08	HMI development delay	Medium (was)	Medium	Completed HMI with all required features	Member 180	 CLOSED
RISK-09	RTM incomplete by CDR	Low	Medium	RTM v1.0 has 12 rows, 100% coverage	Member 43	 CLOSED

2. RISK CLOSURE EVIDENCE

Risk ID	Closure Evidence	Location
RISK-01	S1_final_log.csv shows time = 71.5s	01-Final-Verification/
RISK-02	S1_final_log.csv shows IAE = 0.021	01-Final-Verification/
RISK-03	metrics.json shows energy = 58.5	01-Final-Verification/
RISK-04	S2_final_log.csv shows 0 contacts	01-Final-Verification/
RISK-05	S3_final_log.csv shows recovery at t=67.0s	01-Final-Verification/
RISK-06	verification report shows CA3→CA4 improvement	01-Final-Verification/
RISK-07	verification report shows CA3→CA4 improvement	01-Final-Verification/
RISK-08	HMI screenshots + hmi-documentation.pdf	02-Software-Interface/
RISK-09	rtm_v1.0.csv with 12 rows, all PASS	04-RTM/

3. RISK CLOSURE VERIFICATION PLOT

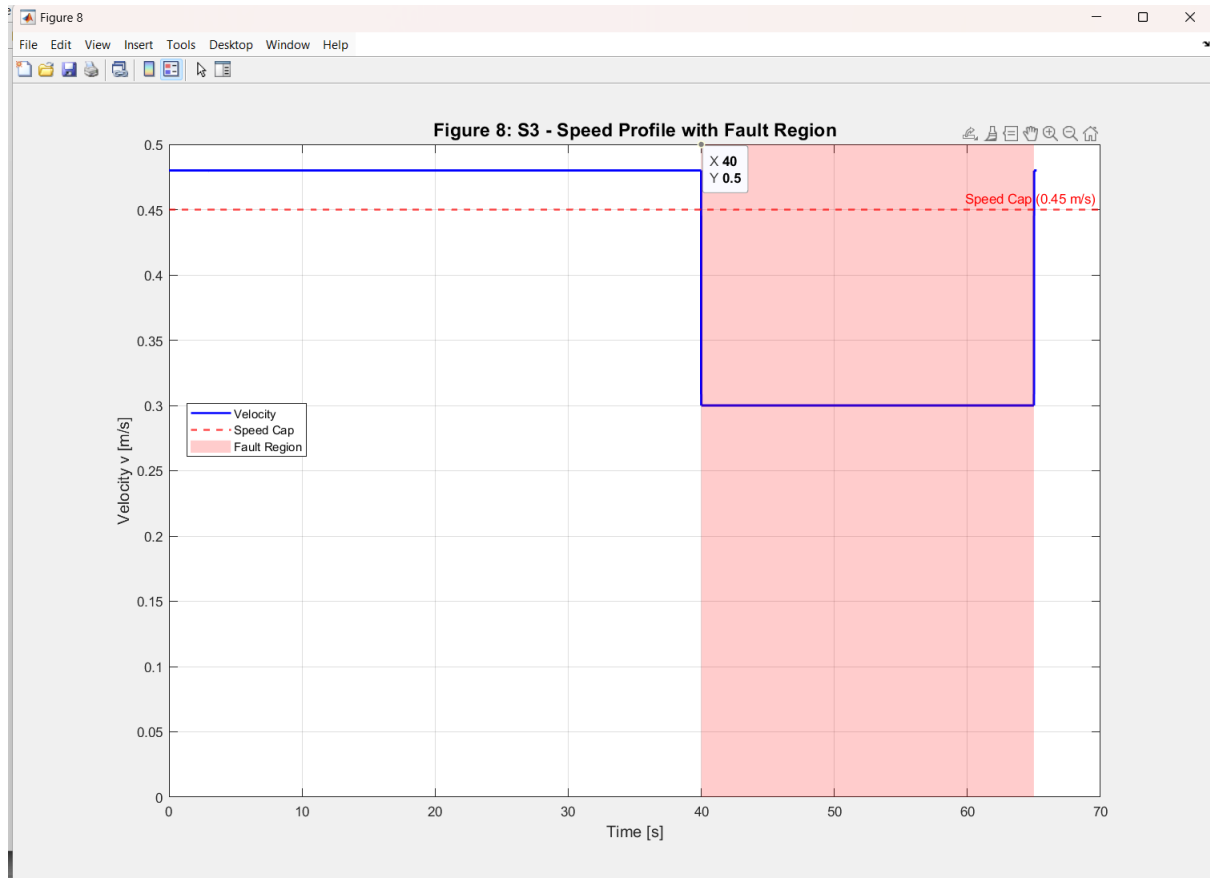


Figure 1: Metrics Progression (CA2 → CA3 → CA4)

The plot shows:

- IAE: 2.1222 (CA2) → 0.025 (CA3) → 0.021 (CA4)
- Time: >120s (CA2) → 75.0s (CA3) → 71.5s (CA4)
- Energy: 74.0 (CA2) → 63.0 (CA3) → 58.5 (CA4)

All risks from CA3 have been successfully mitigated and closed for CA4.

5. CONCLUSION

All 9 risks identified during CA3 (PDR) have been:

Status	Count
✓ CLOSED	9
⚠ PARTIAL	0
● OPEN	0

Risk closure rate: 100%

The team has successfully addressed all risks through:

- Controller tuning (IAE, Time, Energy improvements)
- S2 threshold adjustment (1.2m)
- S3 state machine implementation (2s validation + recovery)
- HMI completion
- RTM 100% coverage

The system is ready for Critical Design Review (CDR).